

FIBONACCI REGENERATION IN A BERNOULLI-RESET GOLDEN RANDOM SHIFT: EXACT QUENCHED-ANNEALED GAP AND GAUSSIAN FLUCTUATIONS

ANONYMOUS

ABSTRACT. At every time, choose independently between the golden-mean adjacency matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{and the reset matrix} \quad E = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix},$$

using E with probability p . The number N_n of fibre paths through n random constraints is a random matrix product. The rank-one identity $EA^kE = F_{k+2}E$ turns its logarithm into a renewal-reward process. We obtain, for $0 < p < 1$,

$$h_q(p) = p^2 \sum_{k \geq 0} (1-p)^k \log F_{k+2}, \quad h_a(p) = \log \frac{1 + \sqrt{5-4p}}{2},$$

and prove the strict disorder gap $h_q(p) < h_a(p)$. We also prove

$$\frac{\log N_n - nh_q(p)}{\sqrt{n}} \implies \mathcal{N}(0, \sigma_p^2),$$

where

$$\sigma_p^2 = p^2 \sum_{k \geq 0} (1-p)^k (\log F_{k+2} - (k+1)h_q(p))^2 > 0.$$

The general random-subshift, random-matrix, and regenerative limit theories are established background. This note records an exact two-matrix specialization and makes no absolute novelty or priority claim.

1. INTRODUCTION

The topological entropy of a deterministic shift of finite type is the logarithm of a Perron eigenvalue. In a random shift, the corresponding path count is a product of adjacency matrices, and two distinct growth rates appear: the almost-sure, or quenched, exponent of the path count and the annealed exponent of its expectation. Products of random matrices have a classical general theory beginning with Furstenberg and Kesten [3]; random subshifts and their thermodynamic formalism are developed, at much greater generality, by Kifer and by Denker–Kifer–Stadlbauer [2, 4].

This note isolates a degenerate but exactly solvable Bernoulli model. One matrix is the familiar presentation of the golden-mean shift [5]. The other deletes its second row and acts as a rank-one reset. A block of ordinary golden constraints between two resets contributes exactly one Fibonacci number. Reset gaps are geometric, so the entire cocycle becomes regenerative rather than merely subadditive.

The resulting calculation has three features. First, it gives a closed series for the quenched fibre entropy. Second, the annealed exponent remains a quadratic Perron root, and a cycle-normalized strict Jensen argument proves that the two exponents differ at every nontrivial reset probability. Third, the same cycles yield a central limit theorem with an explicit positive variance. No perturbation, irreducibility theorem for random operators, or numerical Lyapunov-exponent approximation is required.

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All logarithms are natural. We use the classical Fibonacci normalization $F_0 = 0$, $F_1 = 1$, and $F_{j+2} = F_{j+1} + F_j$, and write $\varphi = (1 + \sqrt{5})/2$. The external-release status of this internal manuscript is **HOLD**: the broad mechanisms have clear owners, and the exact conjunction of formulas below is presented without a claim that it has not appeared elsewhere.

2. THE RANDOM NEAREST-NEIGHBOR SHIFT

Let $\Omega = \{0, 1\}^{\mathbb{Z}}$ and let ν_p be the Bernoulli measure for which $\nu_p\{\omega : \omega_i = 1\} = p$. The symbol 1 marks a reset. Put

$$(1) \quad A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad M_i(\omega) = \begin{cases} A, & \omega_i = 0, \\ E, & \omega_i = 1. \end{cases}$$

For a fixed environment ω , define the fibre

$$(2) \quad X_\omega = \left\{ x \in \{0, 1\}^{\mathbb{Z}} : M_i(\omega)_{x_i, x_{i+1}} = 1 \text{ for every } i \in \mathbb{Z} \right\}.$$

The all-zero point belongs to every fibre, so no exceptional empty fibre arises. The number of paths across the constraints at times $0, \dots, n-1$ is

$$(3) \quad N_n(\omega) = \mathbf{1}^\top M_0(\omega) M_1(\omega) \cdots M_{n-1}(\omega) \mathbf{1}, \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Thus N_n counts words $x_0 \cdots x_n$, not words of length n ; this one-symbol convention does not affect any exponential rate.

When the limits exist, define

$$(4) \quad h_q(p) = \lim_{n \rightarrow \infty} \frac{1}{n} \log N_n(\omega) \quad (\nu_p\text{-a.s.}), \quad h_a(p) = \lim_{n \rightarrow \infty} \frac{1}{n} \log \mathbb{E}_{\nu_p} N_n.$$

The first is the fibre topological entropy along a typical environment; the second averages path counts before taking the logarithm.

3. FIBONACCI REGENERATION AND QUENCHED ENTROPY

The exact solvability is contained in one matrix identity.

Lemma 3.1 (Fibonacci reset identity). *For every $k \geq 0$,*

$$(5) \quad EA^k E = F_{k+2} E.$$

More precisely, if the reset positions in $\{0, \dots, n-1\}$ are $r_1 < \dots < r_m$, put

$$\ell = r_1, \quad q = n-1-r_m, \quad k_j = r_{j+1} - r_j - 1 \quad (1 \leq j < m).$$

Then

$$(6) \quad N_n(\omega) = B_{\ell, q} \prod_{j=1}^{m-1} F_{k_j+2}, \quad B_{\ell, q} = \mathbf{1}^\top A^\ell E A^q \mathbf{1}.$$

The boundary factor satisfies

$$(7) \quad 1 \leq B_{\ell, q} \leq 2^{\ell+q+2}.$$

Proof. For $k \geq 1$, induction gives

$$A^k = \begin{pmatrix} F_{k+1} & F_k \\ F_k & F_{k-1} \end{pmatrix}.$$

Multiplication on the left and right by E gives $EA^k E = F_{k+2} E$. At $k = 0$, the same statement is $E^2 = E = F_2 E$. Repeatedly applying this identity to

$$A^\ell E A^{k_1} E \cdots E A^{k_{m-1}} E A^q$$

proves (6). At least the all-zero state path is counted by $B_{\ell, q}$, while there are at most $2^{\ell+q+2}$ binary state words of the required length. This proves (7). $\square$

Let $r = 1 - p$. For $0 < p < 1$, start with the first reset at or after time zero and enumerate later reset times. The numbers K_j of ordinary A -steps strictly between consecutive resets are independent and satisfy

$$(8) \quad \mathbb{P}(K_j = k) = pr^k, \quad k \geq 0.$$

The cycle length and logarithmic reward are

$$(9) \quad L_j = K_j + 1, \quad R_j = \log F_{K_j+2}.$$

They obey $\mathbb{E}L_j = 1/p$ and have moments of every order.

Theorem 3.2 (Exact quenched fibre entropy). *For $0 < p < 1$, the first limit in (4) exists almost surely and in $L^1(\nu_p)$, and*

$$(10) \quad h_q(p) = p^2 \sum_{k=0}^{\infty} r^k \log F_{k+2}.$$

At the deterministic endpoints,

$$(11) \quad h_q(0) = \log \frac{1 + \sqrt{5}}{2}, \quad h_q(1) = 0.$$

Proof. The renewal-reward strong law applied to (8)–(9) gives

$$(12) \quad \frac{\sum_{j < J(n)} R_j}{n} \rightarrow \frac{\mathbb{E}R_0}{\mathbb{E}L_0} = p^2 \sum_{k \geq 0} r^k \log F_{k+2} \quad \text{almost surely,}$$

where $J(n)$ is the number of completed reset-to-reset cycles seen by time n . The series converges because $\log F_{k+2} = O(k + 1)$.

It remains to compare the reward sum with the exact path count. Let H_n be the longest reset-free run inside the environment word $\omega_0 \cdots \omega_{n-1}$. For any $c > 2/|\log r|$, a union bound gives

$$\mathbb{P}(H_n \geq c \log n) \leq nr^{c \log n},$$

and the right side is summable. The Borel–Cantelli lemma therefore gives $H_n = O(\log n)$ almost surely. In (6), the initial reset delay is almost surely finite and the terminal reset-free length is at most H_n . Hence (7) shows that

$$(13) \quad \log N_n - \sum_{j < J(n)} R_j = O(\log n) \quad \text{almost surely.}$$

Combining (12) and (13) proves the almost-sure formula. Since $0 \leq n^{-1} \log N_n \leq (1 + n^{-1}) \log 2$, dominated convergence gives L^1 convergence.

If $p = 0$, then $N_n = \mathbf{1}^\top A^n \mathbf{1} = F_{n+3}$, whose rate is the logarithm of the golden ratio. If $p = 1$, then $E^n = E$ and $N_n = 2$, so the rate is zero. $\square$

4. ANNEALED EXPONENT AND THE STRICT DISORDER GAP

Independence of the environment matrices gives an exact finite-time annealed formula despite their noncommutativity:

$$(14) \quad \mathbb{E}N_n = \mathbf{1}^\top \overline{M}_p^n \mathbf{1}, \quad \overline{M}_p = (1 - p)A + pE = \begin{pmatrix} 1 & 1 \\ r & 0 \end{pmatrix}.$$

Indeed, successively integrating the last independent factor replaces it by its mean, and iteration yields the displayed product.

Theorem 4.1 (Annealed exponent and strict gap). *For every $0 \leq p \leq 1$,*

$$(15) \quad h_a(p) = \log \lambda_p, \quad \lambda_p = \frac{1 + \sqrt{5 - 4p}}{2}.$$

For every nontrivial disorder parameter,

$$(16) \quad h_q(p) < h_a(p), \quad 0 < p < 1.$$

Equality holds at $p = 0$ and $p = 1$.

Proof. The characteristic polynomial of $\overline{M}_p$ is $t^2 - t - r$. For $p < 1$, the matrix is primitive and its Perron root is λ_p in (15); (14) and Perron–Frobenius theory give the exponent. At $p = 1$, the mean matrix is the idempotent E , and the same formula gives $\lambda_1 = 1$.

For strictness, take a gap K with law (8) and define the positive cycle-normalized gain

$$(17) \quad Z = \frac{F_{K+2}}{\lambda_p^{K+1}}.$$

The Fibonacci generating function is

$$(18) \quad \sum_{k \geq 0} F_{k+2} z^k = \frac{1+z}{1-z-z^2}.$$

Because $\lambda_p^2 = \lambda_p + r$, one has $r/\lambda_p = \lambda_p - 1 < \varphi - 1 = \varphi^{-1}$, so substitution of $z = r/\lambda_p$ into the convergent series (18) gives

$$(19) \quad \mathbb{E}Z = \frac{p}{\lambda_p} \frac{1 + r/\lambda_p}{1 - r/\lambda_p - r^2/\lambda_p^2} = 1.$$

For the last equality, after multiplication by λ_p^2 the denominator is $\lambda_p^2 - r\lambda_p - r^2 = p(\lambda_p + r)$, exactly the corresponding numerator. Moreover, Z is not constant: its values at $K = 0$ and $K = 1$ are λ_p^{-1} and $2\lambda_p^{-2}$, respectively, and $1 < \lambda_p < 2$. Strict concavity of the logarithm and (19) now yield

$$h_q(p) - \log \lambda_p = \frac{\mathbb{E} \log F_{K+2} - (\mathbb{E}L) \log \lambda_p}{\mathbb{E}L} = p \mathbb{E} \log Z < p \log \mathbb{E}Z = 0.$$

The endpoint equalities follow from [theorem 3.2](#) and (15). $\square$

The proof identifies the strict gap with within-cycle variability. It is stronger than applying Jensen’s inequality separately at finite n , since it exhibits a nonconstant normalized gain whose strictness survives after division by time.

5. A RENEWAL CENTRAL LIMIT THEOREM

The same exact cycles determine the fluctuations. Regenerative central limit theorems in the required form are classical; see, for example, the renewal and regenerative chapters of Asmussen [1]. We state all moments and the boundary transfer explicitly for this model.

Theorem 5.1 (Gaussian path-count fluctuations). *For every $0 < p < 1$,*

$$(20) \quad \frac{\log N_n - nh_q(p)}{\sqrt{n}} \xrightarrow{d} \mathcal{N}(0, \sigma_p^2),$$

where

$$(21) \quad \sigma_p^2 = p^2 \sum_{k=0}^{\infty} r^k (\log F_{k+2} - (k+1)h_q(p))^2 > 0.$$

The series converges exponentially.

Proof. Write $h = h_q(p)$ and center one cycle by

$$W_j = R_j - hL_j.$$

The pairs (L_j, R_j) are independent and identically distributed, $\mathbb{E}W_j = 0$, and $|W_j| = O(L_j)$. The geometric tail of L_j therefore gives $\mathbb{E}W_j^2 < \infty$. To keep the delayed initial cycle explicit, let $\tau = \min\{t \geq 0 : \omega_t = 1\}$, put $S_m = \sum_{j < m} L_j$, and for $n > \tau$ set

$$J_n = \max\{m \geq 0 : \tau + S_m \leq n - 1\}.$$

Thus J_n counts reset-to-reset cycles completed inside the observation window. Conditional on the almost-surely finite τ , the future cycle pairs are iid with the law in (8)–(9). The delayed renewal-reward central limit theorem therefore gives

$$(22) \quad \frac{\sum_{j < J_n} R_j - h(n-1-\tau)}{\sqrt{n}} \xrightarrow{d} \mathcal{N}\left(0, \frac{\mathbb{E}W_0^2}{\mathbb{E}L_0}\right).$$

Replacing $h(n-1-\tau)$ by hn changes the numerator by the fixed random quantity $h(\tau+1) = o(\sqrt{n})$ almost surely. The terminal unfinished cycle is bounded by the longest reset-free block H_n , and the exact matrix boundary in (13) is $O(\log n)$ almost surely. Both are $o(\sqrt{n})$, so (22) transfers to $\log N_n$.

Finally, using (8) and $\mathbb{E}L_0 = 1/p$,

$$\frac{\mathbb{E}W_0^2}{\mathbb{E}L_0} = p^2 \sum_{k \geq 0} r^k (\log F_{k+2} - (k+1)h)^2,$$

which is (21). Exponential convergence follows because the squared term is $O((k+1)^2)$. Positivity is also exact: if the variance vanished, the positive-probability cases $K=0$ and $K=1$ would give $0 = W(0) = -h$ and $0 = W(1) = \log 2 - 2h$ simultaneously, a contradiction. $\square$

6. EXACT CONTROLS, SCOPE, AND OWNERSHIP BOUNDARY

The accompanying standard-library program is independent of the proofs at the definition level. It enumerates state paths and environments rather than inserting the theorem formulas. Its discrete layer performs 66,787 integer or rational assertions:

- $EA^kE = F_{k+2}E$ and the Fibonacci form of A^k through $k = 50$;
- matrix counts against direct enumeration of every state path for every environment of lengths $0 \leq n \leq 9$;
- the full renewal factorization for every environment of lengths $0 \leq n \leq 15$;
- the exact identity (14) for three rational values of p and all $0 \leq n \leq 16$;
- the rational generating-function reductions used in the strict-gap proof.

Floating-point evaluations are diagnostics only. Representative values are

p	$h_q(p)$	$h_a(p)$	σ_p^2
0.10	0.447641693679	0.452590731810	0.011161475035
0.25	0.392429805910	0.405465108108	0.029582718017
0.50	0.285678394376	0.311905358182	0.057612583213
0.75	0.156807280841	0.188226406460	0.062358024158
0.90	0.066539324973	0.087651818647	0.037293341513

The ownership boundary is deliberate. Furstenberg–Kesten own the general random-matrix-product setting [3]; Kifer and Denker–Kifer–Stadlbauer own broad random-subshift thermodynamic frameworks [2, 4]; and the renewal-reward strong law and central limit theorem are classical regenerative probability [1]. The golden-mean matrix itself is standard [5]. The residual content of this note is the exact Fibonacci-reset specialization that ties those mechanisms together.

A bounded search through 28 August 2026 using combinations of *random golden mean shift*, *Bernoulli reset*, *quenched entropy*, and the exact matrix identity did not identify a direct source for the entire displayed package. That negative search is a collision firewall, not evidence of absolute novelty. In particular, randomness here acts on time-dependent transition constraints, not on substitution images, and the note does not claim a general theorem for random substitutions or arbitrary matrix pairs. External dissemination remains on HOLD pending a separate expert literature and priority review.

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